Final Project - Additional Resources

In the following table you can find some links to some available tools/libraries that you can use for your final project:

Areas	Name of the library	Links
Docker images for the robot and Knowrob	Windows	**Instructions to download the docker image (windows)**
	Ubuntu	**Instructions to download the docker image (ubuntu)**
Tutorials on tiago	Ros+Tiago	Please skip the first tutorial, everything that you need is inside the package /tiago_public_ws <-- inside the docker container (see above) **https://wiki.ros.org/Robots/TIAGo/Tutorials**
Gazebo		**https://classic.gazebosim.org/tutorials?tut=build_world&cat=build_world** Hint: take a look at your assignment_ws, you will find a "world" folder, which contains the model of the gazebo environment we used in our course.
Datasets with semantic annotations	Gibson	**http://svl.stanford.edu/igibson/**
	Querying knowlede base	**https://vib.ai.uni-bremen.de/page/fallschool/chapter3/**
Control	Moveit	**https://moveit.ros.org/**
	Tiago pick_demo	**https://github.com/pal-robotics/tiago_tutorials/tree/noetic-devel/tiago_pick_demo**
Planning	Rosplan	**https://kcl-planning.github.io/ROSPlan/documentation/**
	CRAM	**https://vib.ai.uni-bremen.de/page/fallschool/chapter2/**
	Knowledge Graphs	**https://vib.ai.uni-bremen.de/page/labs/actionable-knowledge-graph-laboratory/**
	LLMs	**https://vib.ai.uni-bremen.de/page/fallschool/chapter5/**
Perception	Yolo	**https://pjreddie.com/darknet/yolo/**

	OpenCV	https://docs.opencv.org/master/df/d65/tutorial_table_of_content_introduction.htm
	Aruco Marker Detection	https://docs.opencv.org/master/d5/dae/tutorial_aruco_detection.html
	Segment Anything Model 2 (SAM2)	https://github.com/facebookresearch/sam2
	RoboKudo	https://robokudo.ai.uni-bremen.de/
Learning	Weka	https://www.cs.waikato.ac.nz/ml/weka/
	Scikit-learn	https://scikit-learn.org/stable/auto_examples/index.html
	RL + Gazebo	https://www.theconstructsim.com/testing-different-openai-rl-algorithms-with-ros-and-gazebo/
	PyTorch	https://pytorch.org
	Tensorflow	https://www.tensorflow.org
Speech recognition	Whisper	https://github.com/openai/whisper
	pocket sphinx	https://cmusphinx.github.io/wiki/